Summary in Graph

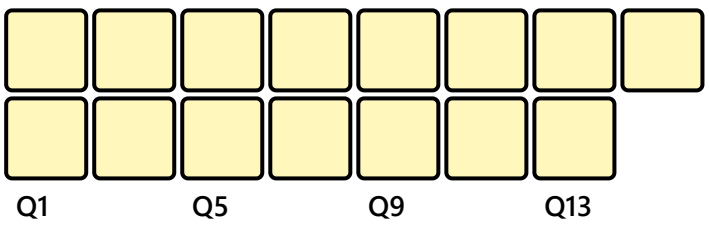
Exam Summary (GO Classes GATE CS | Digital Logic | Number System | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (30) out of 299 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Add the following sets of numbers in $2's$ complement and indicate when an overflow has occurred. Use 4 bits.

A. $2 + 5$

B. $5+7$

C. $4-7$

D. $-4+7$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

What decimal value is represented by the following 8-bit $2's$ complement bit pattern (given in hexadecimal notation)?

$$E1_{16} = ----------_{10}$$

A. 31

B. -31

C. -63

D. 63

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the bit patterns below is the 8-bit $2's$ complement representation of the decimal number 84 ?

A. Not representable

B. 01010100

C. 10101100

D. 00100011

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

In which of the following addition problems (using two's complement notation) does an overflow error occur?

A. $1100+1100$

B. $1100+0100$

C. $0011+1010$

D. $0100+0100$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

A variable that takes thirteen possible values can be communicated using?

A. Thirteen bits.

B. Three bits.

C. $\log_2 13$ bits.

D. Four bits.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Define the value of r in the following: $\sqrt{(41)_r} = (7)_{10}$

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Assign a binary code in some orderly manner to the 52 playing cards. What is the minimum number of bits required?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the answers below corresponds to the sum and overflow result when the following addition is performed using 8-bit unsigned: $10011000 + 1001$?

A. 01111001, with no overflow

B. 10100001, with no overflow

C. 10100001, with overflow

D. 10011001, with no overflow

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the answers below corresponds to the sum and overflow result when the following addition is performed using 8-bit $2's$ complement: $10010000 + 1001$?

A. 10011001, with no overflow

B. 10001001, with overflow

C. 10001001, with no overflow

D. 01111001, with overflow

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Suppose we have two 16-bit 2's complement numbers

$$01X1X0X1XX000XXX$$
$$100XXXX00XXX0X10$$

where some of the bits have not been identified, and they are represented by x. Could the sum of these two numbers possibly result in an overflow? If yes, write 1 in answer. If no, write 0 in answer.

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Perform the following arithmetic operations on 8-bit $2's$ complement values. Express your answer using 8-bit $2's$ complement. Which of the following operations results in an overflow?

A. $00100001 + 00011111$

B. $01101100 + 01011101$

C. $11001100 - 00100101$

D. $10001111 - 01010101$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

What decimal number is represented by the 8-bit $2's$ complement bit pattern (97)H (given in hexadecimal notation)?

A. 10010111

B. 183

C. not representable

D. -105

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type Award: 2 Penalty: 0.67 Digital Logic

Which of the following representations in two's complement notation represents the smallest value?

A. 00000010

B. 11110000

C. 00000001

D. 111111111

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #14 Multiple Choice Type Award: 2 Penalty: 0.67 Digital Logic

Consider the (decimal) number 182, whose binary representation is 10110110. How many positive integers are there in the following set?

$$\{n \in \mathbb{N} : n \leq 182 \text{ and n has } \textit{exactly four} \text{ ones in its binary representation}\}$$

A. 91

B. 70

C. 54

D. 35

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #15 Multiple Select Type Award: 2 Penalty: 0 Digital Logic

Which of the following decimal numbers can be exactly represented in binary notation with a finite number of bits ?

A. 0.1

B. 0.2

C. 0.4

D. 0.5

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

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